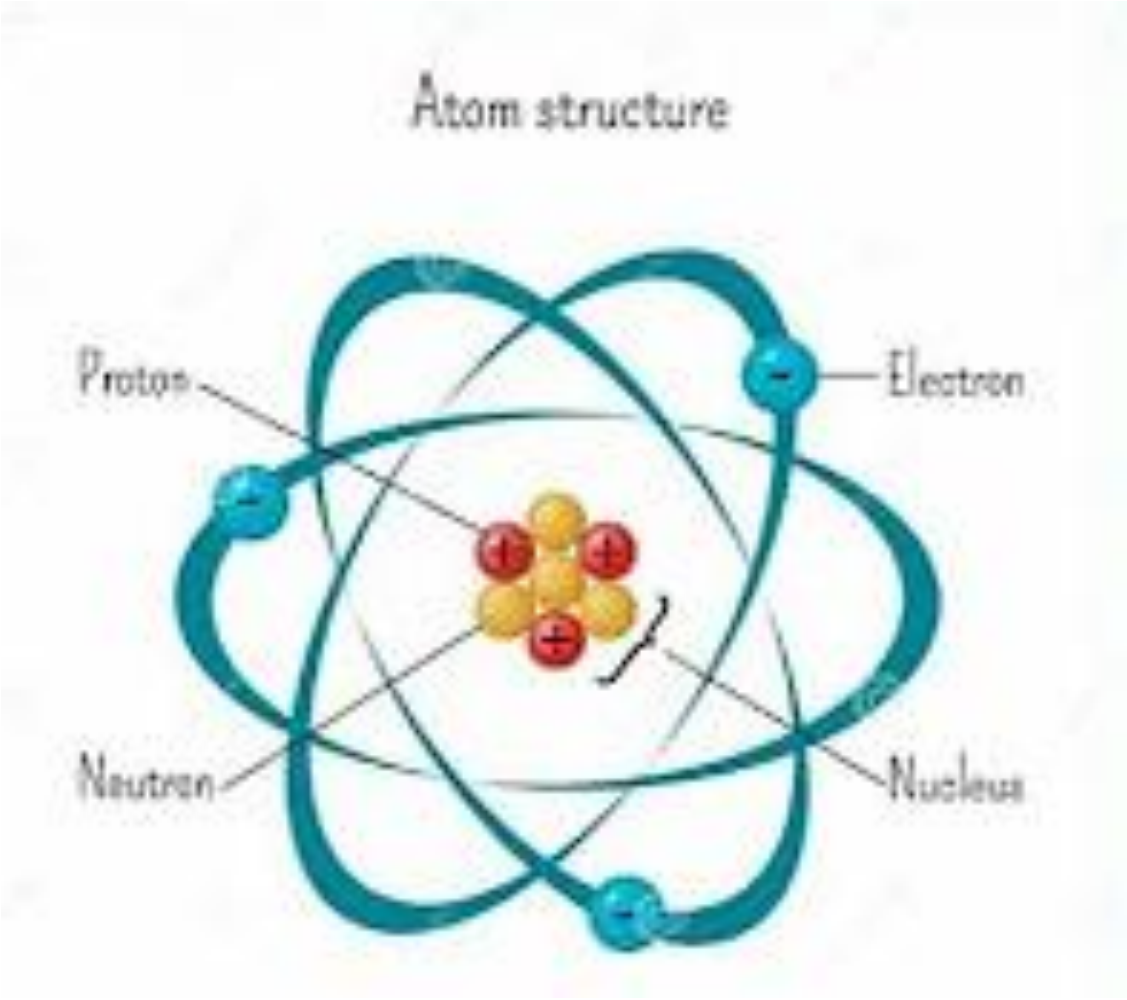
Quantum Mechanical Model of Atoms

- Atom consists of a central, positively charged nucleus surrounded by a number of negatively charged electrons.
- The number of electrons varies from atom to atom.
- The number of electrons is such that their total charge balances the positive charge of the nucleus, thus atom is overall a neutral species.



Wave Function and Probability

- Each electron in a particular energy state is assigned a *wave function* (Ψ).

- The **factors** on which the wave functions depend:

  - Some fundamental physical constants, such as, h, c, π, m_e, e, etc., where h is the Planck's constant, c is the velocity of light, m_e and e are the mass and charge of the electron.
  - Parameters that are peculiar to the system under consideration, expressed in spherical polar coordinates and in Cartesian coordinates.
  - One or more quantum numbers.

- The **probability of finding an electron associated with particular energy state at any point and time** is given by, $\Psi^*\Psi d\tau$, where Ψ^* is the complex conjugate of wave function Ψ and $d\tau$ is the small volume in space to find the probability of an electron.
- In case wave function is real, the equation is $\Psi^2 d\tau$.

Hydrogen Atom

The general form of wave functions for hydrogen atom can be expressed as

$$\Psi_n = f\left(\frac{r}{a_0}\right) e^{-\frac{r}{na_0}} \quad \text{where} \quad a_0 = \frac{h^2}{4\pi^2 m_e e^2}$$

is known as Bohr's radius of first orbit = 0.529 Å

r is the distance of electron from the nucleus in radial coordinates,
$f\left(\frac{r}{a_0}\right)$ is a power series of degree (n-1) and n is the principal quantum number, which can have values as 1, 2, 3, …. ∞.

- Ψ^2 represents the **probability density function** and helps in finding an electron at a given value of r from the nucleus in a state represented by n.

Importance of the Atomic Quantum Numbers

Principal Quantum Number (*n*)

The principal quantum number can only have integral values as 1, 2, 3,..... ∞.
It also governs the size of the electronic orbital.

Azimuthal Quantum Number (*l*)

This is also known as orbital quantum number (l). This quantum number has integral values.
Total number of azimuthal quantum numbers are *n* and they can have values as 0, 1, 2, … (n-1).
This quantum number specifies the number of angular nodes, which is equal to l.

The total number of radial nodes is $n-l-1$. Since the total number of nodes including radial and angular are $n-1$, the total number of angular nodes = $n-1-n+l+1 = l$

It provides information about the **total orbital angular momentum of the electron** as it revolves around the nucleus in its orbital and **shape of the orbital**.

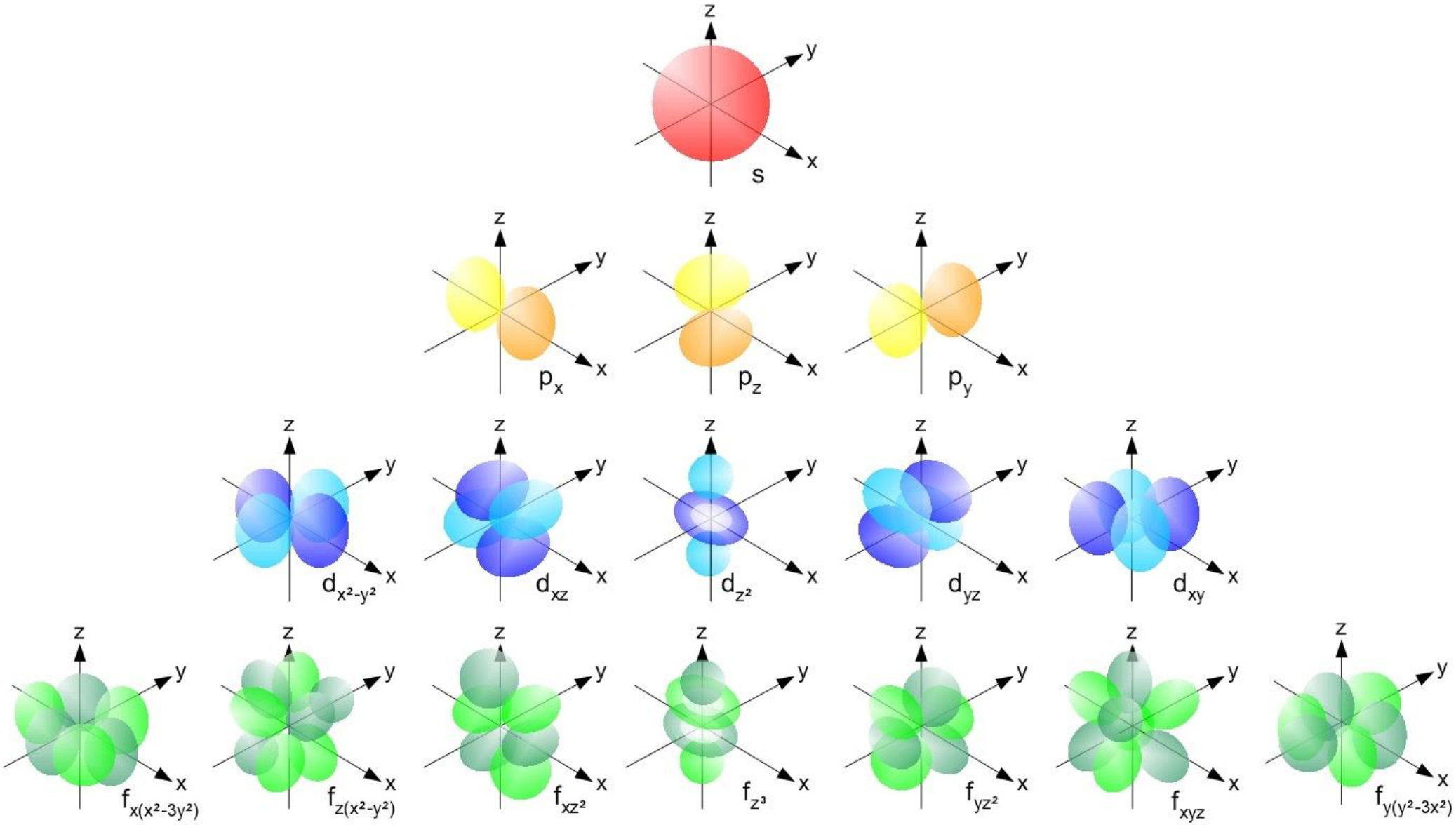
Magnetic Quantum Number (m_l)

This has integral values. 'm_l' is related to *l* by the relation: $l, l-1, l-2, \ldots -(l-1), -l$, i.e., m_l can have $2l+1$ values. m_l can be used to **specify the direction of a particular orbital**.

Spin Quantum Number (*s*)

It measures the **angular momentum due to spin motion of the electron along its own axis**. It can only have a value ½. Spin angular momentum can have two m_s values: +1/2 and -1/2.

Shapes of the orbitals



Fine (Multiplet) Structure of Atomic Terms (States)

Atom (ion) with one outer electron

The **spectral properties** of an atom depend on its **electronic structure** and are determined by the **electronic configuration of the atom** in the ground state.

For a given value of *n*, the number of *l* can be 0, 1, 2, 3 ……. (*n-1)*.

Every state is associated with a <u>definite electronic subgroup</u> with a **maximum number of electrons equal to 2**.

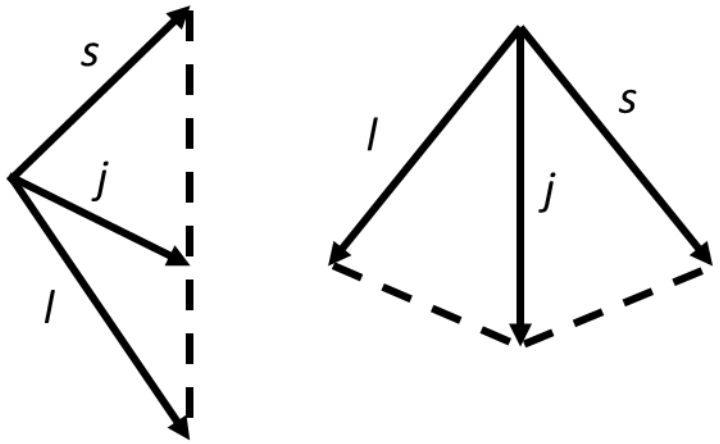
An atom cannot have two or more electrons characterized by the same values of the four independent quantum numbers **(Pauli exclusion principle).**

- The **magnetic interaction of the orbital angular momentum** and the **intrinsic angular momenta of the electrons** (spins) make definite contribution to the <u>atomic state</u>.

- **So, in an atom (ion) with one electron, the interaction of the <u>orbital and the spin angular momentum</u> of this electron needs to be considered.**

- The absolute magnitude of the **resultant angular momentum** gives rise to the expression

$$p = \frac{h}{2\pi}\sqrt{(l \pm s)(l \pm s \pm 1)} \text{ ..... (1)}$$



- In order to characterize the <u>resultant angular momentum</u>, the **<u>total angular momentum quantum number</u> (j)** is introduced. j = *l* ± *s* = *l* ± *1/2*

$$p = \frac{h}{2\pi}\sqrt{j(j+1)} \text{ ..... (2)}$$

The values of the <u>total angular momentum quantum number</u> for various **terms (states)** of <u>hydrogen-like atom (ion)</u> and also for the <u>terms of an atom or ion with one outer electron</u> and with an atomic core having the structure of a noble gas are: (following eqn. 1)

Terms	*l*	*j*			
s	0	1/2			
p	1	1/2	3/2		
d	2		3/2	5/2	
f	3			5/2	7/2

Fine (Multiplet) Structure of Atomic Terms (States)

Atom (ion) with two outer electrons

Four angular momenta - two orbital (l_1 and l_2) and two spin (s_1 and s_2).

Let consider the following two cases:
(i) when the predominant interaction is between vectors l_1 and l_2 and s_1 and s_2
(ii) when the predominant interaction is between vectors l_i and s_i (i = 1, 2)

Case (i): L-S coupling (Russell Saunders coupling)

The vector of the **resultant orbital angular momentum** results due to interaction of vectors l_1 and l_2.

$$\boldsymbol{L} = l_1 + l_2$$

and the vector of the **resultant spin** appears due to the interaction of spin vectors s_1 and s_2.

$$\boldsymbol{S} = s_1 + s_2$$

This type of interaction is called **Russell Saunders coupling**.

The resultant orbital angular momentum $\boldsymbol{L}$ is characterised by the **<u>total orbital angular quantum number</u>** L.

L can take values: $L = l_1 + l_2, l_1 + l_2 - 1, l_1 + l_2 - 2, \ldots \ldots \ldots . |l_1 - l_2|$
depending upon the number of electrons in the atom and the orientation of their orbital vectors.

Thus, **maximum value** of $L = l_1 + l_2$ corresponds to **parallel orientation of the vectors** l_1 and l_2 in the vector model and the **minimum value** of $L = l_1 - l_2$ corresponds to **anti-parallel orientation of the vectors** l_1 and l_2.

The **maximum** and **minimum** values of S can be $S = \frac{1}{2} \pm \frac{1}{2}$, i.e., $S = 1$ or 0.

The resultant $\boldsymbol{L}$ and $\boldsymbol{S}$ then combine to form a resultant $\boldsymbol{J}$ vector which represents the <u>total orbital angular momentum of the whole electron system of the atom</u>.

The **resultant orbital angular momentum vector** $\boldsymbol{J}$ due to the interaction of vectors L and S (L- S coupling) is characterized by the quantum number J, which can take on the values:

$$J = L + S, L + S - 1, L + S - 2, \ldots \ldots . . |L - S|$$

- The number of different values of $J = 2S + 1$ for $L \geq S$ and $J = 2L + 1$ for $L \leq S$.

The number of terms of an atom with two electrons:

The number of S can take up only two values 1 and 0.

So, there are two systems of terms:

(i) **a system of singlets** ($S = 0, J = L$) and (ii) **a system of triplets** ($S = 1, J = L + 1, L, L - 1$).

Terms	L	J				Term symbols
^1S	0	0				$^1S_0$
^1P	1		1			$^1P_1$
^1D	2			2		$^1D_2$
^1F	3				3	$^1F_3$

Singlet terms of an atom with two electrons

Terms	L	J					Term symbols
^3S	0		1				$^3S_1$
^3P	1	0	1	2			$^3P_0$, $^3P_1$, $^3P_2$
^3D	2		1	2	3		$^3D_1$, $^3D_2$, $^3D_3$
^3F	3			2	3	4	$^3F_2$, $^3F_3$, $^3F_4$

Triplet terms of an atom with two electrons

Designation of Atomic Terms

(1) For **atoms or ions with one outer electron**, $l = 0$ and L will also be 0. In such a case, the symbols $s, p, d, f, \ldots$ and $S, P, D, F, \ldots$, denote one and the same thing.
Therefore, the terms of the outer electrons are also the terms of the atom (ion) with one outer electron.

(2) For **an atom (ion) with two electrons**, $l_1 = 1$ and $l_2 = 1$. The resultant $L = 2$.
Thus, the term of an atom with two electrons will be D.

(3) In order to **distinguish the terms belonging to different multiplet systems**, the term is generally written as ^{2S+1}X.

- X stands for the term of an atom (ion). The superscript (2S+1) represents the **multiplicity** of the system corresponding to the term X.
- S is the quantum number that determines the total spin of the atom.

(4) To **differentiate the different components of a multiplet term**, the quantum number J corresponding to the given term X is written as right-hand subscript.

(5) The **most general term of an atom (ion)** is written as $^{2S+1}X_J$

Consider the case $(4s)^2$ $^3S_1$

In this case of single term, J need not be mentioned since the term ^3S like all S terms has $L = 0$ and $J = S$.

However, in the case of a multiplet term, J must be indicated to differentiate the different components.

For the term ^3P, ^3D, etc., we can write ^3P_J, ^3D_J, etc., since 2S+1=3, so $S = 1$ and $L = 1$.
Thus, $J = L + S, L + S - 1, L - S$, i.e., $J = 2,1,0$.

Hence, the components of ^3P term will be $^3P_0$, $^3P_1$ and $^3P_2$.

Similarly, components of ^3D and ^3F terms respectively will be $^3D_1$, $^3D_2$, $^3D_3$ and $^3F_2$, $^3F_3$, $^3F_4$.

In a similar manner, the **doublet P-terms, of alkali elements** ($L = l = 1, J = 3/2, 1/2$) are denoted by symbols *np* $^3P_{3/2}$ and *np* $^3P_{1/2}$ etc.

Case (ii): j-j coupling

This coupling scheme is suitable for heavier elements.

In this scheme, the spin and orbital angular momentum of individual electrons are first coupled to obtain a resultant "spin-orbit" angular momentum for each.

The **corresponding quantum number** is denoted by j such that $j_i = l_i + s_i$ $(i = 1,2, \dots)$

Hence, the total angular momentum vector of the atom, $\boldsymbol{J} = \boldsymbol{j_1} + \boldsymbol{j_2}$
The **total angular momentum vector** $\boldsymbol{J}$ is characterised by the **quantum number** J.

Consider the *sp* configuration of a heavy metal.
For *sp* configuration, we have $l_1 = 0$ and $l_2 = 1$.
and $s_1 = 1/2$ and $s_2 = 1/2$
Hence, $j_1 = l_1 + s_1$ to $|l_1 - s_1|$, $j_1 = 1/2$
and $j_2 = l_2 + s_2$ to $|l_2 - s_2|$, $j_2 = 1 + 1/2$ to $1 - 1/2$ or 3/2 and 1/2
Therefore,
(i) $J = j_1 + j_2$ to $|j_1 - j_2| = \frac{1}{2} + \frac{1}{2}$ to $\left|\frac{1}{2} - \frac{1}{2}\right|$ or 1 and 0
(ii) $J = 2$ and 1